

Steps:

1. Remove Null Productions
2. Remove Unit Productions $\rightarrow$ nonterminal $\rightarrow$ non-terminal
3. Change combination of terminals and/or non-terminals to 2 non-terminals or 1 terminal.

Step 1:

Example,

$$1. S \rightarrow aSa \mid bSb \mid \Lambda$$

$$S \rightarrow aSa$$

$$\Rightarrow a\Lambda a$$

$$\Rightarrow aa$$

$$S \rightarrow bSb$$

$$\Rightarrow b\Lambda b$$

$$\Rightarrow bb$$

Now,

$$S \rightarrow aSa \mid bSb \mid aa \mid bb$$

$$2. S \rightarrow ABAC$$

$$A \rightarrow aA \mid \Lambda$$

$$B \rightarrow bB \mid \Lambda$$

$$C \rightarrow c$$

$$A \rightarrow aA$$

$$\Rightarrow a\Lambda$$

$$\Rightarrow a$$

$$S \rightarrow ABAC$$

$$\Rightarrow \Lambda BAC$$

$$\Rightarrow BAC$$

$$\Rightarrow ABAC$$

$$\Rightarrow ABC$$

$$\Rightarrow \Lambda BAC$$

$$\Rightarrow BC$$

$$B \rightarrow bB$$

$$\Rightarrow b\Lambda$$

$$\Rightarrow b$$

$$S \rightarrow ABAC$$

$$\Rightarrow A\Lambda AC$$

$$\Rightarrow AAC$$

$$\Rightarrow \Lambda\Lambda AC$$

$$\Rightarrow AC$$

$$\Rightarrow \Lambda\Lambda C$$

$$\Rightarrow C$$

Now,

$$S \rightarrow ABAC \mid BAC \mid ABC \mid BC \mid AAC \mid AC \mid C$$

$$A \rightarrow aA \mid a$$

$$B \rightarrow bB \mid b$$

$$C \rightarrow c$$

$$2. S \rightarrow a \mid Xb \mid aYa$$

$$X \rightarrow Y \mid \Lambda$$

$$Y \rightarrow b \mid \Lambda$$

Step 01:

$$S \rightarrow Xb$$

$$\Rightarrow \Lambda b$$

$$\Rightarrow b$$

$$S \rightarrow aYa$$

$$\Rightarrow a\Lambda a$$

$$\Rightarrow aa$$

Step 02:

$$X \rightarrow Y$$

$$\Rightarrow b$$

Now,

$$S \rightarrow a \mid Xb \mid aYa \mid aa \mid b$$

$$X \rightarrow Y$$

$$Y \rightarrow b$$

$$S \rightarrow a \mid Xb \mid aYa \mid aa \mid b$$

$$X \rightarrow a$$

$$Y \rightarrow a$$

Step 2:

Examples,

$$1. S \rightarrow A|bb$$

$$A \rightarrow B|b$$

$$B \rightarrow S|a$$

$$B \rightarrow S$$

$$\Rightarrow bb \Rightarrow A$$

$$\Rightarrow b \Rightarrow a$$

$$S \rightarrow A$$

$$\Rightarrow B \Rightarrow b$$

$$\Rightarrow a$$

$$A \rightarrow B$$

$$\Rightarrow a \Rightarrow S$$

$$\Rightarrow bb \Rightarrow A$$

$$\Rightarrow b$$

Now,

$$S \rightarrow a|bb|b$$

$$A \rightarrow a|bb|b$$

$$B \rightarrow bb|b|a$$

$$2. S \rightarrow XY$$

$$X \rightarrow a$$

$$Y \rightarrow Z|b$$

$$Z \rightarrow M$$

$$M \rightarrow N$$

$$N \rightarrow a$$

$$Y \rightarrow Z$$

$$\Rightarrow M$$

$$\Rightarrow N$$

$$\Rightarrow a$$

$$Z \rightarrow M$$

$$\Rightarrow N$$

$$\Rightarrow a$$

$$M \rightarrow N$$

$$\Rightarrow a$$

$$S \rightarrow XY$$

$$X \rightarrow a$$

$$Y \rightarrow a|b$$

$Z \rightarrow a$
 $M \rightarrow a$
 $N \rightarrow a$

Remove as can never be reached

Step 03:

Examples,

1. $S \rightarrow aSa \mid bSb \mid a \mid b \mid aa \mid bb$

$A \rightarrow a$

$B \rightarrow b$

$S \rightarrow ASA$

$S \rightarrow BSB$

$S \rightarrow AA$

$S \rightarrow BB$

$S \rightarrow AR_1$

$S \rightarrow BR_2$

$R_1 \rightarrow SA$

$R_2 \rightarrow SB$

Now,

$S \rightarrow AR_1 \mid BR_2 \mid a \mid b \mid AA \mid BB$

$R_1 \rightarrow SA$

$R_2 \rightarrow SB$

$A \rightarrow a$

$B \rightarrow b$

2. $S \rightarrow bA \mid aB$

$A \rightarrow bAA \mid aS \mid a$

$B \rightarrow aBB \mid bS \mid b$

$$X \rightarrow a$$

$$Y \rightarrow b$$

$$S \rightarrow YA \mid XB$$

$$A \rightarrow YAA$$

$$A \rightarrow XS$$

$$B \rightarrow XBB$$

$$B \rightarrow YS$$

$$A \rightarrow YR_1$$

$$B \rightarrow XR_2$$

$$R_1 \rightarrow AA$$

$$R_2 \rightarrow BB$$

Now,

$$S \rightarrow YA \mid XB$$

$$A \rightarrow YR_1$$

$$B \rightarrow XR_2$$

$$R_1 \rightarrow AA$$

$$R_2 \rightarrow BB$$

$$X \rightarrow a$$

$$Y \rightarrow b$$

Book Qs:

Q13.

$$i) S \rightarrow X$$

$$X \rightarrow Y$$

$$Y \rightarrow Z$$

$$Z \rightarrow aa$$

$$ii) S \rightarrow SS \mid A$$

$$A \rightarrow SS \mid AS \mid a$$

$$S \rightarrow A$$

$$\Rightarrow SS \Rightarrow AS \Rightarrow a$$

$$S \rightarrow X \quad X \rightarrow Y$$

$$Y \rightarrow Z$$

$$\Rightarrow Y \Rightarrow Z$$

$$\Rightarrow aa$$

$$\Rightarrow Z \Rightarrow aa$$

$$\Rightarrow aa$$

Now,

$$S \rightarrow SS \mid AS \mid a$$

$$A \rightarrow SS \mid AS \mid a$$

$$S \rightarrow aa$$

$$X \rightarrow aa$$

$$Y \rightarrow aa$$

$$Z \rightarrow aa$$

Remove

$$S \rightarrow AA$$

$$A \rightarrow a$$

$$(ii) S \rightarrow aX | bS | a | b$$

$$X \rightarrow aX | a | \Lambda$$

$$S \rightarrow aX$$

$$\Rightarrow a\Lambda$$

$$\Rightarrow a$$

$$X \rightarrow aX$$

$$\Rightarrow a\Lambda$$

$$\Rightarrow a$$

Now,

$$S \rightarrow aX | bS | a | b$$

$$X \rightarrow aX | a$$

Q11.

$$(i) S \rightarrow aX | bX$$

$$X \rightarrow a | b | \Lambda$$

$$S \rightarrow aX$$

$$\Rightarrow a\Lambda$$

$$\Rightarrow a$$

$$S \rightarrow bX$$

$$\Rightarrow b\Lambda$$

$$\Rightarrow b$$

$$S \rightarrow aX | bX | a | b$$

$$X \rightarrow a | b$$

$$(iv) S \rightarrow XaX | bX$$

$$X \rightarrow XaX | XbX | \Lambda$$

$$S \rightarrow XaX$$

$$\Rightarrow \Lambda aX$$

$$\Rightarrow aX$$

$$\Rightarrow Xa\Lambda \Rightarrow \Lambda a\Lambda$$

$$\Rightarrow Xa$$

$$\Rightarrow a$$

$$S \rightarrow bX$$

$$\Rightarrow b\Lambda$$

$$\Rightarrow b$$

Now,

$$S \rightarrow XaX | aX | Xa | a | bX | b$$

$$X \rightarrow XaX | aX | Xa | a | XbX | bX | Xb | b$$

$$(iii) S \rightarrow aS | bX$$

$$X \rightarrow aX | \Lambda$$

$$S \rightarrow bX$$

$$\Rightarrow b\Lambda$$

$$\Rightarrow b$$

$$X \rightarrow aX$$

$$\Rightarrow a\Lambda$$

$$\Rightarrow a$$

$$S \rightarrow aS \mid bX \mid b$$

$$X \rightarrow aX \mid a$$

Q13.

$$(i) S \rightarrow aX \mid Yb$$

$$X \rightarrow S$$

$$Y \rightarrow bY \mid b$$

$$(ii) S \rightarrow AA$$

$$A \rightarrow B \mid BB$$

$$B \rightarrow abB \mid b \mid bb$$

$$X \rightarrow S$$

$$\Rightarrow aX \Rightarrow Yb$$

$$A \rightarrow B$$

$$\Rightarrow abB \Rightarrow b \Rightarrow bb$$

$$S \rightarrow aX \mid Yb$$

$$X \rightarrow aX \mid Yb$$

$$Y \rightarrow bY \mid b$$

Now,

$$S \rightarrow AA$$

$$A \rightarrow abB \mid b \mid bb \mid BB$$

$$B \rightarrow abB \mid b \mid bb$$

$$(iii) S \rightarrow AB$$

$$A \rightarrow B$$

$$B \rightarrow aB \mid Bb \mid \Lambda$$

$$S \rightarrow AB$$

$$\Rightarrow A\Lambda$$

$$\Rightarrow A$$

$$B \rightarrow aB$$

$$\Rightarrow a\Lambda$$

$$\Rightarrow a$$

$$B \rightarrow Bb$$

$$\Rightarrow \Lambda b$$

$$\Rightarrow b$$

$$S \rightarrow AB \mid A$$

$$A \rightarrow B$$

$$B \rightarrow aB \mid Bb \mid A \mid a \mid b$$

$$A \rightarrow B$$

$$\Rightarrow aB \Rightarrow Bb \Rightarrow a \Rightarrow b$$

$$S \rightarrow A$$

$$\Rightarrow B$$

Now,

$$S \rightarrow AB | aB | Bb | a | b$$

$$A \rightarrow aB | Bb | a | b$$

$$B \rightarrow aB | Bb | a | b$$

Q14.

$$(ii) S \rightarrow aSa | SSa | a$$

$$S \rightarrow ASA | SSA | a$$

$$S \rightarrow ASA$$

$$S \rightarrow AR_1 \quad S \rightarrow SR_1$$

$$R_1 \rightarrow SA$$

Now,

$$S \rightarrow AR_1 | SR_1 | a$$

$$R_1 \rightarrow SA$$

$$A \rightarrow a$$

$$(iii) E \rightarrow E + E$$

$$E \rightarrow E * E$$

$$E \rightarrow (E)$$

$$E \rightarrow 7$$

$$E \rightarrow EPE$$

$$E \rightarrow ER_1$$

$$R_1 \rightarrow PE$$

$$E \rightarrow B_1EB_2$$

$$E \rightarrow B_1R_3$$

$$R_3 \rightarrow EB_2$$

$$E \rightarrow EME$$

$$E \rightarrow ER_2$$

$$R_2 \rightarrow ME$$

$$B_1 \rightarrow ($$

$$B_2 \rightarrow)$$

$$P \rightarrow +$$

$$M \rightarrow *$$

Now,

$$E \rightarrow ER_1 | ER_2 | B_1R_3 | 7$$

$$R_1 \rightarrow PE$$

$$R_3 \rightarrow EB_2$$

$$R_2 \rightarrow ME$$

$$B_1 \rightarrow (\quad P \rightarrow +$$

$$B_2 \rightarrow) \quad M \rightarrow *$$

Q1.

$$(iii) \begin{aligned} S &\rightarrow bS \mid \Lambda \mid aA_1 \\ A_1 &\rightarrow aA_1 \mid \Lambda \mid bA_2 \\ A_2 &\rightarrow bS \mid \Lambda \end{aligned}$$

$$(iv) \begin{aligned} S &\rightarrow aS \mid bB_0 \\ B_0 &\rightarrow aB_0 \mid bB_\epsilon \\ B_\epsilon &\rightarrow aS \mid \Lambda \mid bB_0 \end{aligned} \quad (v) \begin{aligned} S &\rightarrow COC \mid a \mid b \\ O &\rightarrow COC \mid a \mid b \\ C &\rightarrow a/b \end{aligned}$$

$$(vi) \begin{aligned} S &\rightarrow Aone \mid Bone \\ Aone &\rightarrow aX \mid Xa \\ Bone &\rightarrow bX \mid Xb \\ X &\rightarrow aX \mid \Lambda \end{aligned}$$

$$(vii) \begin{aligned} S &\rightarrow A_0 \mid B_\epsilon \\ A_0 &\rightarrow bA_0 \mid aA_\epsilon \\ A_\epsilon &\rightarrow bA_\epsilon \mid aA_0 \mid \Lambda \\ B_\epsilon &\rightarrow aB_\epsilon \mid bB_0 \mid \Lambda \\ B_0 &\rightarrow aB_0 \mid bB_\epsilon \end{aligned}$$

Regex to CFG:

$$1. (a+b)^* aa (a+b)^* \quad 2. 0^* 1 (0+1)^*$$

$$\begin{aligned} S &\rightarrow XaaX \\ X &\rightarrow aX \mid bX \mid \Lambda \end{aligned}$$

$$\begin{aligned} S &\rightarrow A1B \\ A &\rightarrow 0A \mid \Lambda \\ B &\rightarrow 0B \mid 1B \mid \Lambda \end{aligned}$$

Q15.

$$(i) \begin{aligned} S &\rightarrow X \\ X &\rightarrow Y \\ Y &\rightarrow Z \\ Z &\rightarrow aa \end{aligned}$$

$$(ii) \begin{aligned} S &\rightarrow SS \mid A \\ A &\rightarrow SS \mid AS \mid a \end{aligned}$$

Step 01: ✓

Step 02:

$$S \rightarrow A$$

$$\Rightarrow SS \Rightarrow AS \Rightarrow a$$

Step 01: ✓

Step 02:

$$S \rightarrow X \quad X \rightarrow Y \quad Y \rightarrow Z$$

$$\Rightarrow Y$$

$$\Rightarrow Z$$

$$\Rightarrow a$$

Now

$$\Rightarrow Z$$

$$\Rightarrow aa$$

$$S \rightarrow SS | AS | a$$

$$\Rightarrow aa$$

$$A \rightarrow SS | AS | a$$

$$S \rightarrow aa$$

$$X \rightarrow aa$$

$$Y \rightarrow aa$$

$$Z \rightarrow aa$$

Step 03:

$$S \rightarrow AA$$

$$X \rightarrow AA$$

$$Y \rightarrow AA$$

$$Z \rightarrow AA$$

$$A \rightarrow a$$

Q17.

$$S \rightarrow AA$$

$$A \rightarrow aB$$

$$B \rightarrow bB | \Lambda$$

abba

$$S \rightarrow AA$$

$$\Rightarrow aBA$$

$$\Rightarrow abbBA$$

$$\Rightarrow abbA$$

$$\Rightarrow abbaB$$

$$\Rightarrow abba$$

$$S \rightarrow SSS | aXb$$

$$X \rightarrow ba | bba | abb$$

$$\begin{array}{c} a \ x \ b \ a \ x \ b \ a \ x \ b \\ \underline{abba} \ \underline{baab} \ \underline{bbab} \end{array}$$

$$S \rightarrow SSS$$

$$\Rightarrow aXbSS$$

$$\Rightarrow abbaBSS$$

$$\Rightarrow abbaB aXbS$$

$\Rightarrow abba baabbbs$

$\Rightarrow abba baabbbaXb$

$\Rightarrow abba baabbbaabbab$